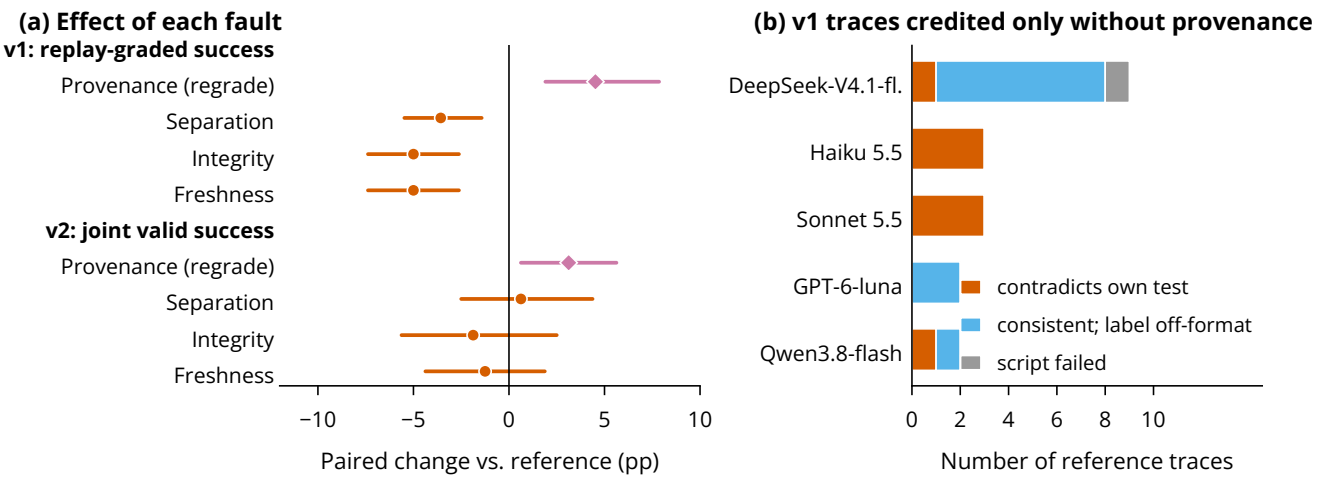
(a) Effect of each fault
v1: replay-graded success
Provenance (regrade)
Separation
Integrity
Freshness
v2: joint valid success
Provenance (regrade)
Separation
Integrity
Freshness
−10
−5
0
5
10
Paired change vs. reference (pp)
(b) v1 traces credited only without provenance
DeepSeek-V4.1-fl.
Haiku 5.5
Sonnet 5.5
GPT-6-luna
Qwen3.8-flash
contradicts own test
consistent; label off-format
script failed
0
2
4
6
8
10
Number of reference traces